SURVEY REPORT

TYPES OF BLOCKCHAINS AND THEIR REAL-TIME USE CASES

Blockchain Technology – Practical BC-4

Practical No.	BC-4
Topic	Survey of blockchain types and applications
Student Name	_____
Class / Division	_____
Roll No.	_____
Academic Year	_____

Department of Computer Engineering

1. Aim

To study the different types of blockchain networks and survey their practical, real-world applications across various industries.

2. Objectives

• Understand the fundamental concept and architecture of blockchain technology.

• Study public, private, consortium and hybrid blockchain networks.

• Compare the characteristics, benefits and limitations of each type.

• Identify real-time use cases in finance, healthcare, supply chains, education, government and other sectors.

• Understand the factors involved in selecting a suitable blockchain platform.

3. Introduction

Blockchain is a type of distributed ledger technology (DLT) in which records are maintained across a network of computers. Information is grouped into blocks, and each block is linked to the previous block using cryptographic techniques. This structure makes unauthorised alteration of recorded information difficult to perform without detection.

Unlike a conventional database controlled by one organisation, a blockchain can allow multiple participants to share and verify records according to agreed rules. Depending on the network design, participants may be able to read data, submit transactions, validate blocks, or perform only selected operations. Blockchain is therefore useful when several parties need a common, auditable record and do not want to rely entirely on one central record keeper.

Blockchain does not automatically make information true or guarantee complete security. The quality of data entered, software design, identity controls, governance and protection of users' private information remain important.

4. Basic Architecture of Blockchain

A typical blockchain system contains the following components:

Block: A data structure containing a group of validated transactions and related metadata.

Transaction: A record of an action, such as transferring a digital asset or updating a shared record.

Cryptographic hash: A fixed-length value generated from data. A block commonly stores the hash of the previous block, linking the chain.

Distributed ledger: A synchronised record shared among participating nodes.

Nodes: Computers or services that participate in the network by storing, validating or relaying information.

Consensus mechanism: A protocol used by network participants to agree on the valid state of the ledger.

Smart contract: Program code deployed on a blockchain that executes defined rules when invoked.

5. Working Principle

Step 1: A participant initiates a transaction using an application or wallet.

Step 2: The transaction is digitally signed and broadcast to the relevant network.

Step 3: Network nodes check it against the protocol rules, such as signature validity and available balance.

Step 4: A consensus process determines which transactions are accepted and in what order.

Step 5: Accepted transactions are grouped into a block and added to the ledger.

Step 6: Participants update their copies of the ledger; the transaction can then be inspected according to network permissions.

6. Types of Blockchain Networks

Blockchain networks are commonly classified according to who can participate, who can validate transactions and how access is governed. The four widely discussed models are public, private, consortium and hybrid blockchains.

6.1 Public Blockchain

A public blockchain is an open network in which anyone can generally read the ledger and, subject to protocol requirements, submit transactions or participate in validation. It is typically decentralised and does not depend on a single organisation to control membership.

Characteristics: Open participation; transparent transaction history; distributed validation; often uses cryptocurrency or economic incentives; governance is generally protocol- and community-based.

Examples: Bitcoin and Ethereum.

Real-time use cases: peer-to-peer cryptocurrency transfers, public token issuance, decentralised finance (DeFi), non-fungible tokens (NFTs), decentralised applications and public verification of digital records.

Advantages: High openness, broad accessibility, transparency and reduced dependence on a single operator.

Limitations: Some networks face throughput constraints, transaction fees and variable confirmation times. Public visibility can also create privacy concerns, and energy use depends on the consensus mechanism.

6.2 Private Blockchain

A private blockchain is operated by one organisation or a clearly defined authority. Participation is permissioned: the operator decides who may join, submit transactions, validate records or access particular information.

Characteristics: Restricted membership; centralised or tightly controlled governance; known participants; configurable access permissions; performance can be tuned for organisational needs.

Examples: Enterprise blockchain deployments using Hyperledger Fabric or other permissioned-ledger platforms. Hyperledger Fabric is a framework, and a particular Fabric network may be configured for one organisation or multiple organisations.

Real-time use cases: internal audit trails, enterprise asset tracking, document workflow, inter-department record sharing and controlled data exchange between business systems.

Advantages: Better control over membership and data access, predictable governance and potentially higher transaction throughput.

Limitations: Less decentralisation, dependence on the governing organisation and the possibility that participants may not trust the operator.

6.3 Consortium Blockchain

A consortium blockchain is governed by a group of organisations rather than one sole operator or an entirely open public community. Selected members operate nodes and share responsibility for validating transactions and maintaining network rules.

Characteristics: Permissioned membership; shared governance; validation by approved organisations; access policies can distinguish between participants.

Examples: Multi-organisation networks built with Hyperledger Fabric or Corda-based enterprise solutions. The exact governance model depends on the deployment.

Real-time use cases: trade finance among banks, logistics and shipping coordination, supply-chain provenance, insurance data exchange and inter-company settlement.

Advantages: Shared control, improved coordination between known parties, and more restricted access than a public network.

Limitations: Governance can be complex; members must agree on rules, responsibilities, upgrades and dispute resolution.

6.4 Hybrid Blockchain

A hybrid blockchain combines selected features of public and private networks. An organisation may keep sensitive information and operational permissions within a controlled environment while publishing proofs, hashes or selected records to a public blockchain for independent verification.

Characteristics: Mix of private and public components; selective transparency; controlled access to confidential data; configurable sharing and verification.

Examples: Architectures that store private business data off-chain or in a permissioned ledger and anchor cryptographic proofs to a public network. Hybrid is an architectural model rather than one single blockchain product.

Real-time use cases: verifiable certificates, land or asset records with public proof and restricted personal details, enterprise supply-chain tracking, and regulated record systems.

Advantages: Balances confidentiality and public verifiability; organisations can choose which information to disclose.

Limitations: Integration and governance are more complicated, and users must understand which records are public, private or externally verifiable.

7. Comparison of Blockchain Types

Feature	Public	Private	Consortium	Hybrid
Access	Open	Restricted	Restricted to member group	Mixed
Governance	Open/community or protocol	One organisation	Multiple organisations	Combination
Validation	Open or protocol-based	Approved operator/nodes	Selected member nodes	Depends on component
Transparency	Generally high	Controlled	Shared among members	Selective
Typical use	Public digital assets	Internal enterprise workflows	Inter-company processes	Selective public verification

8. Real-World Applications of Blockchain

8.1 Banking and Finance

Blockchain can provide a shared record for selected financial transactions and asset movements. Potential applications include cross-border payments, trade-finance documentation, tokenised assets and settlement between institutions. Real deployments must address regulatory compliance, identity checks, privacy and integration with existing payment systems.

8.2 Healthcare

Healthcare organisations can use permissioned ledgers to record access events, share data-consent information and verify the origin of medical records or prescriptions. Patient data should generally remain protected in secure systems; blockchain may store references, permissions or audit proofs rather than exposing full medical records publicly.

8.3 Supply Chain and Logistics

Manufacturers, suppliers, transporters and retailers can share provenance and shipment events. This can help trace products, identify where delays occur and support authenticity checks. The usefulness of the ledger depends on accurate data capture through scanners, sensors and trusted participants.

8.4 Education and Certificates

Institutions can issue digitally verifiable certificates and transcripts. Employers or other institutions can check whether a credential was issued by an authorised institution and whether it has been revoked, reducing manual verification and some forms of certificate fraud.

8.5 Government and Public Records

Possible applications include document verification, public procurement audit trails, land-record workflows and tracking of public-service transactions. Sensitive personal information should not be placed on a public ledger; legal ownership and correction procedures must also be considered.

8.6 Real Estate

Blockchain can support traceable property-document workflows, ownership-history records and tokenisation experiments. It does not by itself establish legal title: reliable links to official land registries, identity checks and legal processes are essential.

8.7 Insurance

Insurers and partners may use shared records to coordinate policy data, claims evidence and settlement. Smart contracts can automate predefined steps when verified conditions are met, but exceptions and disputed claims still require human and legal processes.

8.8 Digital Identity

Blockchain-related identity systems can help users present verifiable credentials issued by trusted organisations. Good designs minimise the personal data placed on-chain and allow credentials to be updated, expired or revoked.

8.9 Internet of Things (IoT)

IoT devices can generate events about equipment status, temperature or location. A blockchain can provide a shared audit trail for selected events across organisations. Device security, data volume, latency and the reliability of sensor inputs remain important design concerns.

8.10 Intellectual Property and Digital Content

Creators may use blockchain timestamps, hashes or token systems to help demonstrate that a digital file or transaction existed at a particular time. Such records can support evidence and licensing workflows, but they do not automatically prove authorship or replace copyright registration and legal procedures.

9. Advantages of Blockchain

- Shared records can reduce repeated reconciliation between participating organisations.
- Cryptographic linking and consensus can make unauthorised record changes easier to detect.
- An auditable transaction history can improve traceability.

• Smart contracts can automate well-defined business rules.

• Distributed operation can reduce reliance on one central database operator in suitable settings.

• Permissioned networks can provide fine-grained access controls for known participants.

10. Limitations and Challenges

• Scalability: transaction throughput and confirmation time may be limited by the network design.

• Cost: infrastructure, transaction fees, integration and ongoing governance can be expensive.

• Privacy: replicated records and transparent transaction histories require careful data protection.

• Regulation: legal requirements differ across countries and industries.

• Interoperability: connecting separate blockchains and legacy systems can be difficult.

• Data quality: incorrect information entered into a ledger can remain difficult to correct.

• Key management: loss or theft of cryptographic keys can cause serious access and security problems.

• Governance: participants need clear rules for upgrades, disputes, permissions and accountability.

11. Blockchain vs. Traditional Database

Aspect	Traditional database	Blockchain
Control	Usually managed by one organisation	Shared among network participants, depending on design
Record changes	Can be updated or deleted by authorised administrators	Records are typically append-oriented; corrections are added as new transactions
Trust model	Trust in database operator and controls	Trust in protocol, participants and governance
Performance	Often high for centralised workloads	Depends on consensus and network architecture
Best fit	Applications with a clear trusted owner	Multi-party workflows needing shared auditability

12. Popular Blockchain Platforms

Bitcoin: A public blockchain designed primarily for peer-to-peer digital value transfer.

Ethereum: A public blockchain platform that supports smart contracts and decentralised applications.

Hyperledger Fabric: An open-source framework for building permissioned enterprise blockchain networks with modular components and membership controls.

R3 Corda: A distributed-ledger platform designed for business workflows in which participants share relevant data on a need-to-know basis.

Polygon: A blockchain ecosystem focused on scaling and interoperability with Ethereum-compatible applications.

13. Factors for Selecting a Blockchain

• Determine whether participants are public, known organisations or members of one organisation.

• Identify which data must be public, private or selectively disclosed.

• Evaluate transaction volume, speed, cost and availability requirements.

• Define who validates transactions and who governs upgrades or disputes.

• Check applicable laws, data-protection rules and audit requirements.

• Assess integration with existing databases, applications and identity systems.

• Use blockchain only when shared control, traceability or tamper-evident records provide a clear benefit over a conventional database.

14. Future Scope

Blockchain development is moving towards more efficient consensus mechanisms, improved privacy, interoperability between networks, tokenisation of real-world assets and integration with artificial intelligence and IoT systems. Enterprise adoption is likely to focus on specific workflows where multiple parties need a common source of truth. The long-term success of blockchain will depend not only on technical improvements but also on regulation, usability, security and sound governance.

15. Conclusion

Blockchain is a distributed-ledger technology that can provide shared, traceable and tamper-evident records. Public, private, consortium and hybrid blockchains differ mainly in access, governance, validation and transparency. Public networks are suitable for open participation and public digital assets, while private and consortium networks are often used for controlled enterprise collaboration. Hybrid designs can combine private data handling with public verification. Applications are being explored in finance, healthcare, logistics, education, government, insurance, identity and digital content. However, blockchain is not a universal replacement for databases; it should be selected only when its shared-ledger properties solve a genuine business or technical problem.

16. References

1. National Informatics Centre, Government of India, Centre of Excellence in Blockchain Technology: https://blockchain.gov.in/

2. National Institute of Standards and Technology (NIST), Blockchain: https://www.nist.gov/information-technology/blockchain

3. Hyperledger Fabric Documentation – What is Hyperledger Fabric?: https://hyperledger-fabric.readthedocs.io/en/latest/whatis.html

4. Ethereum Developer Documentation: https://ethereum.org/en/developers/docs/

5. ISO, ISO/TR 3242:2022 – Blockchain and distributed ledger technologies — Use cases: https://www.iso.org/standard/79543.html

6. NITI Aayog, Blockchain: The India Strategy: https://www.niti.gov.in/node/296

Result: The survey of major blockchain types and their real-world use cases was completed. The characteristics, advantages, limitations and application areas of public, private, consortium and hybrid blockchains were studied.